Pulse Width Modulation

Introduction

In the realm of power electronics, efficient and controlled energy conversion is paramount. Previous discussions focused on single-phase AC to DC converters, such as boost and flyback-based Power Factor Correction (PFC) converters, which are typically employed for low-power applications, generally up to 7 kilowatts. These converters, operating in Continuous Conduction Mode (CCM) or Discontinuous Conduction Mode (DCM), are crucial for regulating output voltage and achieving unity power factor. However, for power levels exceeding 7 kilowatts, the demand shifts towards three-phase AC to DC converters, which necessitate more sophisticated control strategies.

This lecture introduces Pulse Width Modulation (PWM) as a foundational technique for implementing three-phase AC to DC conversion. PWM is a versatile modulation method that enables precise control over the output voltage, ensuring a constant DC output while maintaining a unity power factor current drawn from the AC source. Understanding PWM is essential for designing robust and efficient power conversion systems that meet the demands of higher power applications.

Core Concepts

Term	Definition	Significance
`Half-bridge`	A fundamental building block of voltage source converters, comprising two switches (S1 and S2) connected in series across a DC link.	It forms the basic unit for generating modulated voltage waveforms, crucial for AC-DC and DC-AC conversion.
`Pole Voltage (VAO)`	The voltage at the midpoint (pole) between the two switches of a half-bridge, referenced to a fictitious midpoint 'O' of the DC link.	Its controlled variation, achieved through switching, is the essence of PWM to synthesize desired output waveforms.
`Carrier Signal`	A high-frequency, repetitive waveform (e.g., triangle or ramp) used as a reference for comparison in PWM.	Its frequency determines the switching frequency of the converter, influencing

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		harmonic content and filter size.
`Modulating Signal`	A low-frequency signal (DC or sinusoidal) that dictates the desired output waveform, compared against the carrier signal.	Its amplitude and frequency determine the amplitude and fundamental frequency of the converter's output voltage.
`Pulse Width Modulation (PWM)`	A technique where the width of voltage pulses is varied to control the average output voltage of a converter.	It allows for the generation of variable average voltages from a fixed DC source, enabling AC synthesis or controlled DC output.
`Duty Ratio (D)`	The fraction of a switching period during which a switch is ON, or the proportion of time a specific voltage level is applied.	Directly controls the average voltage applied to the load, and in PWM, it is varied to shape the output waveform.
`Modulation Index (m)`	The ratio of the peak amplitude of the modulating signal (Vm) to the peak amplitude of the carrier signal (Vc).	It quantifies the depth of modulation and directly influences the amplitude of the fundamental component of the output voltage.
`Harmonics`	Unwanted frequency components present in the output voltage or current waveform, in addition to the desired fundamental frequency.	They represent power quality issues; PWM aims to push them to higher frequencies for easier filtering.
`Switching Frequency (fc)`	The frequency at which the power switches in the converter are turned ON and OFF, typically determined by the carrier frequency.	A higher switching frequency allows for smaller filter components and better harmonic suppression.

Detailed Analysis

1. Recap and Transition to Three-Phase Converters

The lecture began by revisiting single-phase AC to DC converters, specifically boost and flyback-based PFC converters. These systems were analyzed for their operation, design considerations, small-signal modeling, and closed-loop control, focusing on maintaining output voltage regulation and achieving unity power factor. Both Continuous Conduction Mode (CCM) and Discontinuous Conduction Mode (DCM) operations were discussed, highlighting their distinct characteristics and control implications. Notably, DCM operation in buck-boost derived converters (like flyback) can achieve a voltage follower mode, presenting a resistive input impedance and thus drawing unity power factor current. While effective for low-power applications (up to 7 kW), these single-phase solutions are insufficient for higher power demands, necessitating a shift to three-phase AC to DC converters. This transition introduces the need for advanced modulation techniques like Pulse Width Modulation (PWM).

2. The Half-Bridge Configuration: A Fundamental Building Block

The half-bridge configuration serves as the fundamental building block for voltage source converters, whether AC-DC, DC-AC, or DC-DC. It consists of two switches, S1 (top) and S2 (bottom), connected in series across a DC voltage source. These switches are typically realized using devices like MOSFETs or IGBTs, each with a freewheeling diode, enabling them to block voltage of one polarity and allow current flow in either direction.

The half-bridge operates in two primary states:
* **State A**: S1 is ON, and S2 is OFF. In this state, the pole voltage `VAO` (voltage at the midpoint 'A' with respect to the fictitious DC link midpoint 'O') is `Vdc/2`. The voltage `VAN` (voltage at 'A' with respect to the negative DC link terminal 'N') is `Vdc`.
* **State B**: S1 is OFF, and S2 is ON. Here, `VAO` is `-Vdc/2`, and `VAN` is `0`.

By rapidly switching between these two states, the half-bridge can impose a voltage at its pole that varies between `Vdc/2` and `-Vdc/2`. This ability to control the pole voltage is central to PWM.

3. Principle of Pulse Width Modulation (PWM)

Pulse Width Modulation is a technique used to control the output voltage of a converter by varying the duration (width) of the voltage pulses at the half-bridge's

pole. Instead of changing the magnitude of the applied voltage (`Vdc/2` or `-Vdc/2`), PWM alters the time for which each voltage level is applied.

The core idea involves comparing a high-frequency **carrier signal** (typically a triangle or ramp waveform) with a low-frequency **modulating signal** (also known as a reference signal).
* The carrier signal has a constant amplitude (e.g., `+Vc` to `-Vc`) and a high frequency (`fc`).
* The modulating signal can be a DC quantity or a low-frequency AC signal (`fm`), where `fm` is significantly less than `fc` . This ensures that the modulating signal appears relatively constant over one carrier cycle.

The switching logic is straightforward:
* If the modulating wave is greater than the carrier wave, the top switch S1 is turned ON, applying `Vdc/2` to the pole.
* Otherwise (modulating wave is less than the carrier wave), the bottom switch S2 is turned ON, applying `-Vdc/2` to the pole.

This comparison generates a series of pulses at the pole, where the width of these pulses is directly proportional to the instantaneous value of the modulating signal.

4. Derivation of Average Pole Voltage Variation

To understand how PWM controls the output, we analyze the average pole voltage `VAO_av` over one carrier cycle (`Tc`). Considering a half-cycle of the carrier (`Tc/2`) due to symmetry, the average voltage is calculated by integrating the applied voltages over their respective durations.

When S1 is ON, `Vdc/2` is applied for a duration `t_on` . When S2 is ON, `-Vdc/2` is applied for `(Tc/2 - t_on)` .
The average pole voltage over `Tc/2` is:
`VAO_av = [(-Vdc/2) * (Tc/2 - t_on) + (Vdc/2) * t_on] / (Tc/2)`
Simplifying this expression yields:
`VAO_av = -Vdc/2 + Vdc * (t_on / (Tc/2))`

We define the **duty ratio (D)** as `D = t_on / (Tc/2)` . Substituting this into the equation:
`VAO_av = -Vdc/2 + Vdc * D`

The duty ratio `D` can be related to the modulating and carrier signal amplitudes. By comparing the slopes of the carrier and modulating signals at the switching instant, it can be derived that:
`D = (Vm / 2Vc) + 1/2` (Equation 1)

Substituting this expression for `D` back into the average pole voltage equation:
`VAO_av = -Vdc/2 + Vdc * [(Vm / 2Vc) + 1/2]`

`VA0_av = -Vdc/2 + (Vm * Vdc / 2Vc) + Vdc/2`
`VA0_av = (Vm / 2Vc) * Vdc`

This crucial result shows that the average pole voltage variation over a carrier cycle is directly proportional to the amplitude of the modulating signal (`Vm`). This proportionality allows precise control of the output voltage by simply adjusting the modulating signal.

5. Sinusoidal Pulse Width Modulation (SPWM)

If the modulating signal is made to vary sinusoidally, i.e., `Vm(t) = Vm * sin(ωt)` , where `ω = 2πfm` and `fm` is the fundamental frequency, then the average pole voltage will also vary sinusoidally:
`VA0_av(t) = (Vm * sin(ωt) / 2Vc) * Vdc`

This technique is known as Sinusoidal Pulse Width Modulation (SPWM) because the width of the pole voltage pulses varies sinusoidally over the line cycle. SPWM is widely used in inverters to generate AC voltage from a DC bus, or conversely, in rectifiers to convert AC to DC.

A key parameter in SPWM is the **modulation index (m)**, defined as the ratio of the peak of the modulating signal (`Vm`) to the peak of the carrier signal (`Vc`):
`m = Vm / Vc`

Substituting `m` into the `VA0_av` equation:
`VA0_av(t) = (m * Vdc / 2) * sin(ωt)`

This expression clearly shows that the fundamental component of the pole voltage is directly proportional to the modulation index `m` and the DC link voltage `Vdc` . By controlling `m` , the amplitude of the fundamental output voltage can be precisely adjusted.

6. Harmonics in SPWM and Filtering Considerations

While SPWM effectively generates a fundamental sinusoidal component, the instantaneous pole voltage is still a series of pulses switching between `Vdc/2` and `-Vdc/2` . This pulsed nature inherently introduces **harmonics** into the output waveform, in addition to the desired fundamental frequency.

The harmonic content of an SPWM modulated half-bridge output can be mathematically described using a double Fourier series. The harmonics are primarily present at the carrier frequency and its sidebands, given by the general formula:
`h = i * mf ± j`
where `mf` is the **frequency modulation ratio** (`fc / fm`), `i` is an integer (1, 2, 3...), and `j` is an integer (0, 1, 2, 3...).
The rules for `i` and `j` are:

* If `i` is odd, then `j` must be even (e.g., `i=1, j=0, 2, 4, 6...`).
* If `i` is even, then `j` must be odd (e.g., `i=2, j=1, 3, 5...`).

For example, with `fm = 50 Hz` and `mf = 21` (meaning `fc = 1050 Hz`), harmonics would appear at:
* Fundamental: `50 Hz`
* Sidebands around `mf`: `(1 * 21) ± 0 = 21` (i.e., `1050 Hz`), `(1 * 21) ± 2 = 19, 23`, `(1 * 21) ± 4 = 17, 25`, etc.
* Sidebands around `2*mf`: `(2 * 21) ± 1 = 41, 43`, `(2 * 21) ± 3 = 39, 45`, etc.

A simulation example with `Vdc = 400V`, `m = 0.9`, and `mf = 21` shows a fundamental component of `180V` (`0.9 * 400 / 2`). The harmonic spectrum reveals that the magnitude of harmonics generally decreases as the harmonic order increases.

A critical advantage of PWM, particularly with a high carrier frequency, is the ability to push these unwanted harmonics to much higher frequencies, far away from the fundamental component. This allows for the design of smaller and more effective low-pass filters to remove the harmonics, leaving primarily the desired fundamental component. By increasing the switching frequency (`fc`), the harmonics are shifted further up the frequency spectrum, simplifying filter design and reducing filter size and cost.

Key Takeaways

- [ ] Single-phase AC-DC converters are suitable for low power (<7kW), while three-phase converters are needed for higher power applications.
- [ ] Pulse Width Modulation (PWM) is a key technique for three-phase AC-DC conversion, enabling constant output voltage and unity power factor.
- [ ] The half-bridge configuration, comprising two switches, is the fundamental building block for voltage source converters.
- [ ] PWM controls output voltage by varying the width of pole voltage pulses, not their magnitude.
- [ ] PWM operates by comparing a high-frequency carrier signal with a low-frequency modulating signal to determine switch ON/OFF states.
- [ ] The average pole voltage is directly proportional to the amplitude of the modulating signal.
- [ ] Sinusoidal Pulse Width Modulation (SPWM) uses a sinusoidal modulating signal to produce a sinusoidal fundamental output voltage.
- [ ] The modulation index (m) directly controls the amplitude of the fundamental output voltage component.

- [ ] Harmonics are generated at the carrier frequency and its sidebands, but a high carrier frequency pushes them to higher frequencies, simplifying filtering.

Conclusion

Pulse Width Modulation stands as a cornerstone technique in modern power electronics, particularly for high-power applications like three-phase AC to DC conversion. By precisely controlling the duration of voltage pulses, PWM allows for the synthesis of desired output waveforms from a fixed DC link, ensuring both voltage regulation and power factor correction. The ability to vary the average pole voltage proportionally to a modulating signal, especially in Sinusoidal PWM, provides a robust mechanism for generating clean AC or DC outputs.

The strategic use of a high-frequency carrier signal is crucial, as it enables the displacement of unwanted harmonics to higher frequencies, where they can be efficiently filtered out using smaller and more cost-effective passive components. This principle is vital for achieving high power quality and efficiency in power conversion systems. As we move forward, the concepts of PWM will be instrumental in understanding and stabilizing the design and operation of three-phase AC to DC power converters, forming the basis for advanced control strategies in charging infrastructure and other critical applications.